Some Potentially Useful Formulas and Definitions

$$z = \frac{(\hat{p}_1 - \hat{p}_2)}{\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}} \qquad t = \frac{(\bar{x}_1 - \bar{x}_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \qquad t = \frac{\bar{x}_d}{\left(\frac{s_d}{\sqrt{n}}\right)} \qquad \bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

- Calculate the mean of the absolute values of the pairwise differences in sample proportions
- The p-value is the probability of observing something as or more extreme as what we observed, under the null.

p-value	Strength of Evidence
0.10 < p-value	Not much evidence against the null hypothesis; null is plausible
0.05 < p-value ≤ 0.10	Moderate evidence against the null hypothesis
0.01 < p-value ≤ 0.05	Strong evidence against the null hypothesis
p-value ≤ 0.01	Very strong evidence against the null hypothesis

Table 1: Guidelines for evaluating the strength of evidence from a p-value